CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 14: Equation of a Line

Previous year board questions, topic-wise: 11 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Forms of the equation

1. $ABCD$ is a parallelogram with vertices $A(3,3)$, $B(0,-2)$, $C(-4,-2)$ and $D(-1,3)$, read from a graph. (a) Find the point P where the diagonals meet. (b) Find the slopes of CB and DA and check that these sides are parallel. (c) Find the equation of the diagonal AC. ICSE 2025, 5 marks

2. Triangle ABC has vertices $A(1,-1)$, $B(0,4)$ and $C(-6,4)$, and D is the mid-point of BC. Find (a) the coordinates of D (b) the equation of the median AD. ICSE 2023, 3 marks

3. $A(2,5)$, $B(-1,2)$ and $C(5,8)$ are the vertices of a triangle. M is the point on AB with $AM : MB = 1:2$. Find M, and then the equation of the line through C and M. ICSE 2018, 4 marks

2 Lines parallel or perpendicular to a given line

4. Which line passes through the origin and is parallel to $3x+4y+7=0$? (a) $3x+4y+5=0$ (b) $4x-3y-5=0$ (c) $4x-3y=0$ (d) $3x+4y=0$ ICSE 2025, 1 mark

5. The line PQ meets the positive x-axis at P, 5 units from the origin, and the negative y-axis at Q, 3 units from the origin. The line RS passes through the origin and is perpendicular to PQ. Find (a) the coordinates of P and Q (b) the equation of RS. ICSE 2025, 3 marks

6. (a) The lines $kx-y+4=0$ and $2y=6x+7$ are perpendicular. Find k. (b) Find the equation of the line parallel to $2y=6x+7$ that passes through $(-1,1)$. ICSE 2024, 3 marks

7. The lines $5x-3y+2=0$ and $6x-py+7=0$ are perpendicular. Find p. Then find the equation of the line through $(-2,-1)$ parallel to $6x-py+7=0$. ICSE 2020, 3 marks

8. The vertices of $\triangle ABC$ are $A(3,8)$, $B(-1,2)$ and $C(6,-6)$. Find (a) the slope of BC (b) the equation of the line through A perpendicular to BC. ICSE 2019, 3 marks

9. The lines $3x-5y=7$ and $4x+ay+9=0$ are perpendicular. Find a. ICSE 2018, 3 marks

10. $A(-1,3)$, $B(4,2)$ and $C(3,-2)$ are the vertices of a triangle. Find (a) the centroid G (b) the equation of the line through G parallel to AC. ICSE 2017, 3 marks

3 Inclination and slope

11. The points $A(x,y)$, $B(3,-2)$ and $C(4,-5)$ are collinear. Write y in terms of x. (a) $3x-11$ (b) $11-3x$ (c) $3x-7$ (d) $7-3x$ ICSE 2024, 1 mark

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